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Department of Electrical Engineering
END-TERM EXAMINATION, November 2024
B.Tech. III Semester

Subject: EMEC-I
Max. Marks: 50

Subject Code: EE-212
Time: 3 Hours

Note: All Questions are compulsory. All the parts of each question should be written at one place.

Q.No.	Question	Marks	CO
1. a.	With the help of suitable connection and phasor diagrams, show that the power factor of a balanced three phase load can be determined using two wattmeters.	5 M	1
1. b.	A two-wattmeter method is used to measure the power of a three-phase delta connected load. The wattmeter readings are $W_1 = 800W$ and $W_2 = -400W$. The line voltage is 400 V and line current is 5 A. Calculate (a) the total power consumed (b) apparent power of the system (c) reactive power of the load (d) Load impedance (e) power factor of the load.	5 M	1
2. a.	The efficiency of a 20 kVA, 2500/250 V, 1-phase transformer at upf is 98% at rated load and at half rated load. Determine (a) the transformer core loss and ohmic losses (b) percentage value of the equivalent resistance of the transformer?	5 M	5
2.b.	Explain with necessary connection and phasor diagrams how three phase supply can be converted into a balanced two-phase supply. If the load is balanced on one side, show that it will be balanced on the other side.	5 M	4
3. a.	Derive the relationship between primary and secondary quantities (voltage and current) for standard 3-phase transformer connections with a neat diagram? Consider. 'V' and 'I' are input line voltage and line current, and 'a' is phase turns ratio.	5 M	4

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3.b.	A 500 kVA load at 0.8 pf lagging is supplied by three 1-phase transformers connected in Δ-Δ. Each of the Δ-Δ transformers is rated at 200 kVA, 2500/250 V. If one defective transformer is removed from service, calculate for the V-V connection: (a) kVA load carried by each transformer (b) Percentage rated load carried by each transformer (c) Total kVA rating of the transformer bank in V-V (d) Ratio of V-V bank to Δ-Δ bank transformer ratings (e) Percentage increase in load on each transformer when one transformer is removed	5 M	4
4.a.	A 8-pole, DC generator has armature winding containing 32 coils of 6 turns each. Its flux per pole is 0.06 Wb. The machine is running at 250 rpm. Calculate the pole pitch, and the induced armature voltage for (i) simplex lap wound (ii) simplex wave wound?	5 M	2
4.b.	A 250 V compound generator has armature, series field, and shunt field resistances of 0.4 Ω , 0.2 Ω , and 125 Ω respectively. If this generator supplies 10 kW at rated voltage, find the emf generated in the armature when the machine is connected (a) long shunt (b) short shunt. Ignore armature reaction and allow 1 volt per brush for contact drop.	5 M	2
5.a.	Explain armature reaction and its effect on dc machine. Sketch the distribution of flux in air gap due to (i) Main field mmf only (ii) Armature mmf only (iii) Both main field and armature mmf.	5 M	2
5.b.	A shunt-wound motor with an armature resistance of 0.2 ohms is connected across a 400 V supply. The armature current is 40 A and the speed of the motor is 1000 rpm. Calculate the additional resistance which should be connected in series with the armature to reduce its speed to 700 rpm. Assume that the armature current remains the same.	5 M	3